



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to R.N.K FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

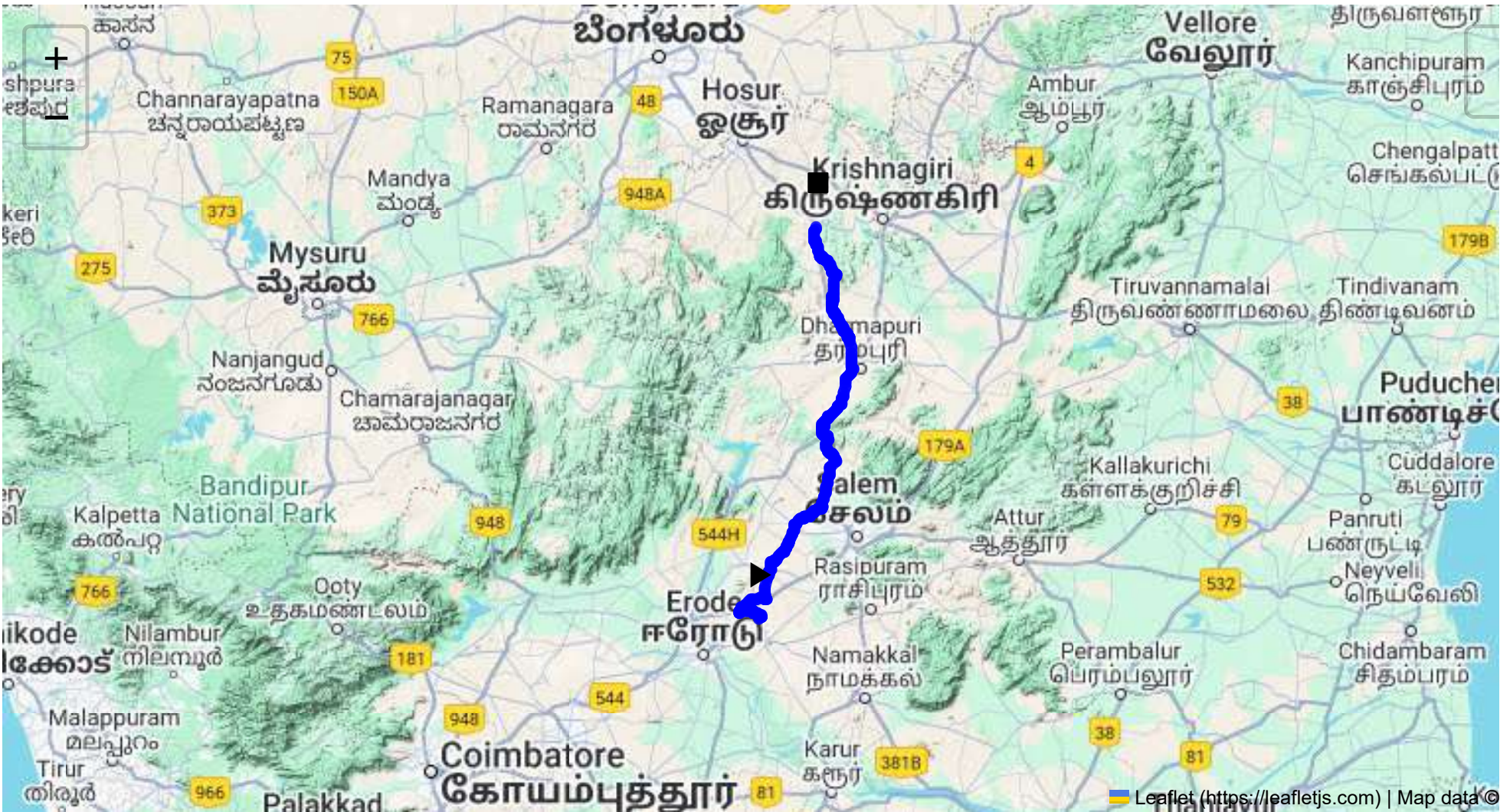
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 155.50 km**  
**Estimated Duration: 3.4 hours**  
**Adjusted Duration (Heavy Vehicle): 4.3 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.503987, 78.031157)**



Welcome to the Journey Risk Management Study

To provide a comprehensive safety analysis of the given route from Sangagiri to Rayakottai via multiple waypoints, we must consider various aspects that can impact the transit of heavy vehicles, especially those carrying hazardous materials. Below is a detailed report based on your requirements:

1. Overview of the Route Map:



- The route stretches approximately 155.50 kilometers, starting from Sangagiri and concluding in Rayakottai, passing through areas like Sankari, Salem, Dharmapuri, and Palacode.
- It involves a combination of state highways (SH 86 and SH 17) and main roads, traversing urban, semi-urban, and rural regions.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The region experiences a tropical climate with significant rainfall during the southwest monsoon (June to September) and the northeast monsoon (October to December).
- Potential weather hazards include heavy rainfall leading to waterlogging, reduced visibility, and slippery roads, especially during monsoon seasons.

## 3. Analysis of Traffic Patterns:

- Urban areas like Salem and proximal regions can experience peak hour congestion, usually in the morning (8 AM - 10 AM) and late afternoon (5 PM - 7 PM).
- Smaller towns along the route may see less congestion but can have localized busy periods, often around market areas or roadway junctions.

## 4. Assessment of Road Quality and Infrastructure:

- State highways are generally well-maintained, but local roads might present issues like potholes or uneven surfaces.
- Bridges and underpasses should be reviewed for weight limitations that might be pertinent to heavy or oversized vehicles.

## 5. Suggestions for Alternative Routes for Emergencies:

- An alternate route could involve using NH 44 (part of India's Golden Quadrilateral) for faster and potentially safer transit, though it may require deviations.
- The National Highways Authority of India (NHAI) app or local Google Maps traffic layer can offer real-time rerouting suggestions during transit.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport:

- The transport of hazardous materials in Tamil Nadu is governed by the Motor Vehicles Act and several state-specific guidelines. Compliance with mandatory signage, documentation, and permits is crucial.
- Night travel for hazardous materials might be restricted, requiring permits.

## 7. Overview of Historical Incidents:

- No significant publicized incidents specific to hazardous material transport were reported historically along this route, but caution and preparedness for possible emergencies are advised.

## 8. Environmental Considerations and Sensitive Areas:

- The route passes near agricultural land and small water bodies, hence risk assessment should include potential spill impacts on local ecosystems.
- Avoidance of sensitive zones like schools and hospitals during peak hours is recommended.

## 9. Analysis of Communication Coverage:

- While major towns should have strong cellular and data coverage, rural segments might have sporadic dead zones.

- It’s advisable for drivers to have alternate communication methods, like satellite phones, for emergencies.

10. **Estimated Emergency Response Times:**

- Urban areas: 15-30 minutes, thanks to proximity to emergency services.
- Rural areas: Could be 30-60 minutes depending on road conditions and the remoteness.
- Local coordination with police and fire stations en route is advised for planning and quicker response.

11. **Overall Summary of Risk Assessment:**

- While the primary route is feasible, attention must be paid to monsoon-related hazards, potential road quality issues, local traffic regulations, and emergency preparedness.
- Training with a focus on defensive driving, familiarity with the route, and emergency drills can mitigate potential risks significantly.
- Continuous monitoring of traffic and weather updates will be essential to maintain a high safety standard for transporting hazardous materials.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45764, 77.85219 | 30 KM/Hr    |
| 5  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 6  | Turn       | High       | 11.45571, 77.81702 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.45654, 77.81706 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.46916, 77.82431 | 30 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.49225, 77.89560 | 10 KM/Hr    |
| 10 | Turn       | High       | 11.49453, 77.88488 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.56348, 77.90077 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.57547, 77.90567 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.59189, 77.90645 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.59323, 77.90796 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.63857, 77.94831 | 30 KM/Hr    |
| 16 | Turn       | High       | 11.64207, 77.95199 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 17 | Turn       | Medium     | 11.64708, 77.95119 | 30 KM/Hr    |
| 18 | Turn       | High       | 11.69596, 77.97122 | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.71721, 77.99982 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.71923, 78.01140 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.73772, 78.03606 | 30 KM/Hr    |
| 22 | Turn       | High       | 11.73665, 78.03907 | 15 KM/Hr    |
| 23 | Turn       | Medium     | 11.74226, 78.04688 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.77961, 78.05775 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.86873, 78.08374 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.87966, 78.08187 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.91211, 78.05852 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.92609, 78.06513 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.93334, 78.06409 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.93719, 78.05372 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.95457, 78.04685 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 11.95688, 78.04803 | 30 KM/Hr    |
| 33 | Turn       | High       | 11.95828, 78.04983 | 15 KM/Hr    |
| 34 | Turn       | Medium     | 11.95997, 78.04946 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 11.96335, 78.05141 | 30 KM/Hr    |
| 36 | Turn       | High       | 11.96408, 78.05131 | 15 KM/Hr    |
| 37 | Turn       | Medium     | 11.96511, 78.04966 | 30 KM/Hr    |
| 38 | Turn       | High       | 11.96608, 78.04919 | 15 KM/Hr    |
| 39 | Turn       | Medium     | 11.96819, 78.05088 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 12.12010, 78.13282 | 30 KM/Hr    |
| 41 | Turn       | High       | 12.20914, 78.11681 | 15 KM/Hr    |
| 42 | Blind Spot | Blind Spot | 12.20970, 78.11680 | 10 KM/Hr    |
| 43 | Blind Spot | Blind Spot | 12.20378, 78.12304 | 10 KM/Hr    |
| 44 | Turn       | Medium     | 12.22451, 78.10903 | 30 KM/Hr    |
| 45 | Turn       | Medium     | 12.23481, 78.10507 | 30 KM/Hr    |
| 46 | Turn       | Medium     | 12.23511, 78.10444 | 30 KM/Hr    |
| 47 | Turn       | Medium     | 12.26532, 78.08427 | 30 KM/Hr    |
| 48 | Turn       | High       | 12.29602, 78.07407 | 15 KM/Hr    |
| 49 | Turn       | Medium     | 12.29641, 78.07527 | 30 KM/Hr    |
| 50 | Turn       | High       | 12.31180, 78.07854 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 51 | Blind Spot | Blind Spot | 12.31360, 78.07464 | 10 KM/Hr    |
| 52 | Turn       | High       | 12.37241, 78.08518 | 15 KM/Hr    |
| 53 | Turn       | Medium     | 12.37573, 78.08362 | 30 KM/Hr    |
| 54 | Turn       | High       | 12.37875, 78.08503 | 15 KM/Hr    |
| 55 | Turn       | Medium     | 12.38703, 78.07477 | 30 KM/Hr    |
| 56 | Turn       | Medium     | 12.39347, 78.07169 | 30 KM/Hr    |
| 57 | Turn       | Medium     | 12.40605, 78.06717 | 30 KM/Hr    |
| 58 | Turn       | Medium     | 12.41316, 78.06064 | 30 KM/Hr    |
| 59 | Turn       | Medium     | 12.41690, 78.05592 | 30 KM/Hr    |
| 60 | Turn       | Medium     | 12.41829, 78.05120 | 30 KM/Hr    |
| 61 | Blind Spot | Blind Spot | 12.42547, 78.04645 | 10 KM/Hr    |
| 62 | Turn       | High       | 12.42472, 78.04656 | 15 KM/Hr    |
| 63 | Turn       | Medium     | 12.42292, 78.04507 | 30 KM/Hr    |
| 64 | Turn       | Medium     | 12.42274, 78.04450 | 30 KM/Hr    |
| 65 | Blind Spot | Blind Spot | 12.42350, 78.04307 | 10 KM/Hr    |
| 66 | Turn       | Medium     | 12.42376, 78.04341 | 30 KM/Hr    |
| 67 | Turn       | Medium     | 12.42677, 78.04418 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                       | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------|------------------------|-------------|------------|
| 1  | hospital | Sri Shanthi Hospital       | 11.569272, 77.901937   | 30 km/h     | Medium     |
| 2  | clinic   | Sun Eye Care               | 11.6936242, 77.9730948 | 30 km/h     | Medium     |
| 3  | hospital | Dr. Pandiyarajan Hospital  | 11.692704, 77.969344   | 30 km/h     | Medium     |
| 4  | hospital | Dr. Sengodan Hospital      | 11.6959527, 77.9695325 | 30 km/h     | Medium     |
| 5  | hospital | Omalur Government Hospital | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 6  | clinic   | R.R. Nursing Home          | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 7  | hospital | Sri Hari Hospital          | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 8  | hospital | Sugumar Hospital           | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 9  | hospital | Sri Sugam Hospital         | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 10 | hospital | Saravana Hospital          | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 11 | hospital | Arun Hospital              | 11.7429892, 78.0466881 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 12 | clinic   | Saratha Clinic                     | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 13 | clinic   | Sri Meenakshi Nursing Home, Salem  | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 14 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 15 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 17 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 18 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 19 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 20 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 21 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 22 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 23 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 24 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 25 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 26 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 28 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 29 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 30 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |
| 32 | hospital | Government Hospital Palacode       | 12.2964824, 78.0761735 | 30 km/h     | Medium     |

## Crowded Spots

|    | type    | name                                   | coordinates            | speed_limit | risk_level |
|----|---------|----------------------------------------|------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School             | 11.4546193, 77.8142445 | 30 km/h     | Medium     |
| 16 | school  | Fathima Hr Sec School                  | 11.7410508, 78.0479272 | 30 km/h     | Medium     |
| 27 | college | Dhirajlal Gandhi College of Technology | 11.7968014, 78.0562095 | 30 km/h     | Medium     |
| 31 | college | Laxminarayana Polytechnic College      | 12.0421277, 78.103203  | 30 km/h     | Medium     |

## Route Photos of Risky Spots





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44900, 77.87406**





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.45352, 77.85698**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45764, 77.85219**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.46303, 77.84945**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45571, 77.81702



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45654, 77.81706





Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49453, 77.88488





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56348, 77.90077



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57547, 77.90567





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.59189, 77.90645**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.59323, 77.90796**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.63857, 77.94831**





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.64207, 77.95199



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64708, 77.95119





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.69596, 77.97122



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71721, 77.99982



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71923, 78.01140





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73772, 78.03606



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.73665, 78.03907





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.74226, 78.04688**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.77961, 78.05775**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.86873, 78.08374**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87966, 78.08187



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91211, 78.05852





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92609, 78.06513



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93334, 78.06409



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93719, 78.05372





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95457, 78.04685



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95688, 78.04803





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95828, 78.04983



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95997, 78.04946



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96335, 78.05141





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96408, 78.05131



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96511, 78.04966





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96608, 78.04919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96819, 78.05088



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12010, 78.13282





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.20914, 78.11681



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.20970, 78.11680





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.22451, 78.10903



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.23511, 78.10444



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.26532, 78.08427





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.29602, 78.07407



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.29641, 78.07527





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.31180, 78.07854



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.31360, 78.07464



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.37241, 78.08518





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.37573, 78.08362



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.37875, 78.08503





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.38703, 78.07477



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.39347, 78.07169



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.40605, 78.06717





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.41316, 78.06064



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.41690, 78.05592





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.41829, 78.05120



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.42547, 78.04645



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.42472, 78.04656





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.42292, 78.04507



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.42274, 78.04450



